

HALDIA INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTION UNDER MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL)

Paper Code : 001-PCC-CSE405-N

Paper Name : Formal Language and Automata Theory

Time Allotted : 3 Hours

Full Marks : 70

*The figures in the margin indicate full marks**Candidates are required to give their answers in their own words as far as practicable***Group-A****(Multiple Choice Type Questions)**

1. Choose the correct alternatives from the followings:

[15 x 1 = 15]

(I) A language is decidable if:

CO-1

- (a) TM halts on all inputs
- (b) TM loops forever
- (c) TM rejects all inputs
- (d) TM accepts some inputs

(II) If NFA has n states, DFA can have at most:

CO-2

- (a) n
- (b) n^2
- (c) 2^n
- (d) $n!$

(III) A language is Turing recognizable if:

CO-4

- (a) TM accepts it
- (b) TM always rejects
- (c) TM always halts
- (d) No TM exists

(IV) CFG belongs to:

CO-1

- (a) Type-1
- (b) Type-2
- (c) Type-3
- (d) Type-0

(V) A parse tree represents:

CO-4

- (a) Final string
- (b) Structure of derivation
- (c) Alphabet
- (d) Language

(VI) NFA to DFA conversion uses:

CO-2

- (a) Pumping lemma
- (b) Subset construction
- (c) Recursion
- (d) Grammar

(VII) Pumping lemma applies to strings of length:

CO-3

- (a) $< p$
- (b) $= p$
- (c) $\geq p$
- (d) $\leq p$

(VIII) Pumping lemma is used to:

CO-3

- (a) Prove language is regular
- (b) Prove language is not regular

- (c) Convert DFA
- (d) Minimize automata
- (IX) In CFG, production rules are of the form:
- (a) $A \rightarrow aB$
- (b) $A \rightarrow \alpha$
- (c) $\alpha \rightarrow \beta$ ✗
- (d) $AB \rightarrow a$ ✗
- (X) Context-Free Grammar is defined as:
- (a) (Q, Σ, δ)
- (b) (V, T, P, S)
- (c) (Σ, L)
- (d) (P, Q)
- (XI) A regular expression is used to describe:
- (a) Context-free languages
- (b) Regular languages
- (c) Context-sensitive languages
- (d) All languages
- (XII) The tape in a Turing Machine is:
- (a) Finite
- (b) Infinite
- (c) Circular
- (d) Empty
- (XIII) Regular grammar corresponds to:
- (a) DFA
- (b) NFA
- (c) Both DFA and NFA
- (d) PDA
- (XIV) Regular expressions and finite automata are:
- (a) Not equivalent
- (b) Equivalent
- (c) Partially equivalent
- (d) Unrelated
- (XV) A DFA is defined by how many components?
- (a) 3
- (b) 4
- (c) 5
- (d) 6

CO-1

CO-2

CO-3

CO-4

CO-2

CO-3

CO-1

Group-B

(Short Answer Type Questions)

[3 x 5 = 15]

2(I) Explain string and language. Generate some strings by using this Language $L = \{ W1 \mid W \in \{0,1\}^* \}$

[5] CO-1

OR

2(II) Differentiate between Terminal and Non-Terminal symbols.

[5] CO-1

3(I) Define DFA and NFA with example.

[5] CO-2

OR

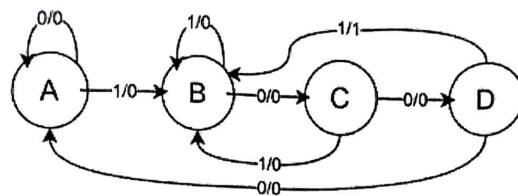
3(II) Construct a DFA by using the Language $L = \{ W \in \{0,1\}^* \mid \#a(W) \bmod 3 \}$

[5] CO-2

4(I) Convert the Mealy machine to a Moore machine.

[5] CO-3

!!



OR

4(II) Prove that $L = \{a^n b^n \mid n \geq 0\}$ is not a regular language using Pumping Lemma.

[5] CO-3

Group-C

(Long Answer Type Questions)

[4 x 10 = 40]

5(I) Explain parse trees and their relation to derivations.

[10] CO-1

OR

5(II) (a) Generate some strings by using this Language $L = \{ \text{Set of all strings over } \{0,1\} \text{ containing '00'} \}$

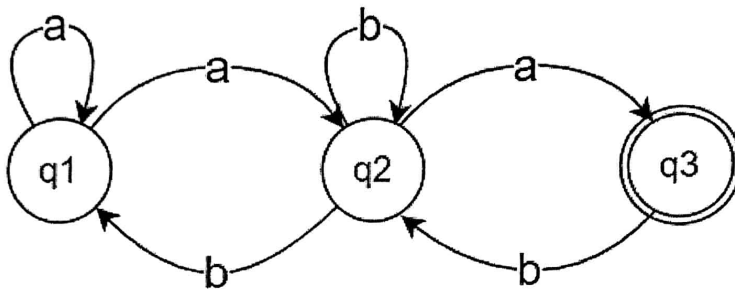
[2] CO-1

(b) Construct a NFA by using the above Language then Convert the NFA to its equivalent DFA.

[8]

6(I) Convert the following NFA to a regular expression.

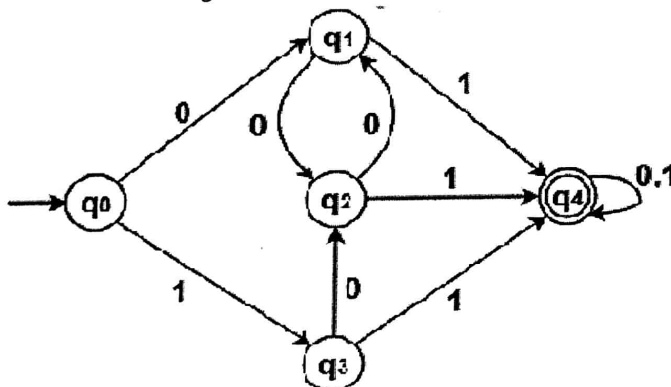
[10] CO-2



OR

6(II) Minimize the following DFA

[10] CO-2



[2] CO-3

[8]

7(I) (a) What are regular expressions?

(b) Prove that $0^*1(0+10^*1)^*$ is equal to $(1+00^*1)+(1+00^*1)(0+10^*1)^*(0+10^*1)$

OR

[10] CO-3

7(II) Design Finite Automata from the given Regular Expressions

1. $a+b$

2. ab

3. $[ab + (b+aa)b^*a]$

[7] CO-4

[3]

8(I) (a) Define Turing Machine with formal definition and diagram.

(b) What is Universal Turing Machine.

OR

[10] CO-4

8(II) Describe variants of Turing Machines and their equivalence.

*** END OF PAPER ***